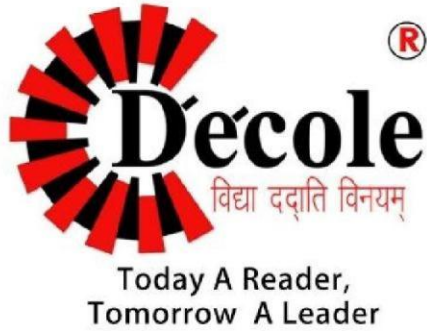




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Title of Project

Assignment File

Subject Name

Data Structure and C++

Submitted By

Jaishree Gahlot

Department - BCA

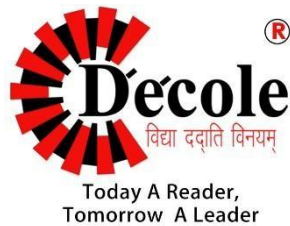
Semester – 2nd

Session – 2025-26

Project Report of
Data Structure and C++

Submitted to
Dezyne École College
Towards the Partial Fulfilment of
BCA 2nd Semester

By
Jaishree Gahlot



www.dezyneecole.com

2024-2025

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We are proud to be part of Dezyne École College, an institution that truly nurtures professionalism, innovation, and purpose.

With sincere appreciation,

Jaishree Gahlot

Department of BCA

Dezyne École College, Ajmer

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Q1. WAP to print all pattern using C++ manipulators:

1024	*	-----***-----
512	***	-----*****-----
64	*****	-----***-----
8	*****	-----*****-----
-----	*****	-----***-----
1608	*****	-----***-----

Code:

```
#include<iostream>
#include<iomanip>
using namespace std
; class Ass1
{
public: void
Pattern1()
{
    int a =
    1024,b=512,c=64,d=8,e;
    e=a+b+c+d;    cout<<"Print
    Pattern 1 --\n\n";
    cout<<setw(5)<<a<<endl;
    cout<<setw(5)<<b<<endl;
    cout<<setw(5)<<c<<endl;
    cout<<setw(5)<<d<<endl;
    cout<<setw(5)<<"-----\n";
    cout<<setw(5)<<e<<endl;
}
void Pattern2 ()
{
    int i,N=5;
    cout<<"Print Patter 2 -- \n\n";
```

```

        for(i=1 ; i<=N ; i++){          cout<<setw(N-
i)<<setfill(' ')<<"";          cout<<setw(i+i-
1)<<setfill('*')<<""<<endl;

        }

    }

    void Pattern3()

    {

        int i,N=5;

        cout<<"Print Pattern 3 -- \n\n";

        for(i=1;i<=N;i++){

            if (i % 2 == 0){

                cout<<setw(3)<<setfill('-')<<"";

                cout<<setw(7)<<setfill('*')<<"";          cout<<setw(3)<<setfill('-
')<<""<<endl;

            }else{

                cout<<setw(5)<<setfill('-')<<"";

                cout<<setw(3)<<setfill('*')<<"";          cout<<setw(5)<<setfill('-
')<<""<<endl;

            }

        }

    }

};

int main()

{

    Ass1 a;

    a.Pattern1();

    a.Pattern2();

    a.Pattern3();

}

```

Output

```
Print Pattern 1 --
1024
 512
  64
   8
-----
1608
Print Patter 2 --
  *
 ***
*****
*****
*****
*****
Print Pattern 3 --
-----xxx-----
-----xxxxx-----
-----xxx-----
-----xxxxx-----
-----xxx-----
```

Q2. Write a program in C++ to perform the following operations using classes.

a. Find the Greatest Common Divisor (GCD) of two numbers given by user.

b. Take a number and perform count and sum of all even and odd numbers and print it.

Code:

```
#include<iostream>

using namespace std;

class Ass2
{
public:
    void GCD()
    {
        cout<<"Greatest Common Divisor ..\n\n";

        int n1 , n2;

        cout<<"Enter Number 1 --";cin>>n1;

        cout<<"Enter Number 2 --";cin>>n2;    int m =
(n1 < n2)?n1:n2;

        int gcd = 0;    for( int i =
1; i<=m ; i++){        if ( n1 % i
==0 && n2 % i==0)
```

```

        gcd = i;
    }

    cout<<"GCD of "<<n1 <<" & "<<n2<<" is -- "<<gcd<<"\n\n";
}

void CSEvenOdd()
{
    cout<<"\n\nCount & Sum of Even and Odd Digit .. \n\n";

    int num,se,ce,so,co;

se=ce=so=co=0;

    cout<<"Enter a Number --";

    cin>>num;

    cout<<"\n\nCount & Sum of Odd and Even Digit in "<<num<<" are -- \n\n";

    for( ;num>0;num=num/10){
        if(num%2==0){
            ce++;

            se=se+(num%10);
        }

        else{
co++;

            so=so+(num%10);
        }
    }

    cout<<"Count of Odd Digits are -- "<<co<<endl;

    cout<<"Sum of Odd Digits are -- "<<so<<endl;

    cout<<"Count of Even Digits are -- "<<ce<<endl;

    cout<<"Sum of Even Digits are -- "<<se<<endl;

    }

};

int main()

```

```
{
    Ass2 a;
    a.GCD();
    a.CSEvenOdd();
}
```

Output

```
Greatest Common Divisor ..
Enter Number 1 --15
Enter Number 2 --65
GCD of 15 & 65 is -- 5

Count & Sum of Even and Odd Digit ..
Enter a Number --123456789

Count & Sum of Odd and Even Digit in 123456789 are --
Count of Odd Digits are -- 5
Sum of Odd Digits are -- 25
Count of Even Digits are -- 4
Sum of Even Digits are -- 20
```

Q3 . WAP to perform the following pattern using class:

```
Take length using user - 6
1 2 3 4 5 6
7 8 9 0 9
0 9 0 9
0 9 0
9 0
9
0 9
0 9 0
9 0 9 0
9 0 9 8 7
6 5 4 3 2 1
```

Code:

```
#include<iostream>

#include<iomanip>

using namespace std;

class Ass3
{
public:
    int N ,i ,j ,C ,S ,P ,PC ,TOT;

    void get()
```

```

{
    C = N = 6;
    S = P = 1;
    PC = 0;
    TOT = (((N*(N+1))/2)*2)-1;
}

void Pattern()
{
    for(i=1;i<=(N+N-1);i++){
cout<<setw(S)<<setfill(' ')<<"";
for(j=1;j<=C;j++,PC++){
    (PC<9 || PC>=(TOT-9))? cout<<P<<" ":(PC % 2 == 1)? cout<<"0 " : cout<<"9
";    P = (PC<9)?++P:(PC>=(TOT-10))?--P:P;
    }
    cout<<"\n";
    C = (i < N)? --C : ++C;
    S = (i < N)? ++S : --S;
    }
}

};

int main()
{
    Ass3 a;
    a.get();
    a.Pattern();
}

Output

```



Q4. Write a program to print the following series:

a. Calculate the sum of the series $(1*1) + (2*2) + (3*3) + (4*4) + (5*5) + \dots + (n*n)$

Sample Output:

Input the value for nth term: 5

$1*1 = 1$

$2*2 = 4$

$3*3 = 9$

$4*4 = 16$

$5*5 = 25$

The sum of the above series is: 55

b. Calculate the series $(1) + (1+2) + (1+2+3) + (1+2+3+4) + \dots + (1+2+3+4+\dots+n)$.

Sample Output:

Input the value for nth term: 5

$1 = 1$

$1+2 = 3$

$1+2+3 = 6$

$1+2+3+4 = 10$

$1+2+3+4+5 = 15$

The sum of the above series is: 35

Code:

```
#include<iostream>
```

```
#include<conio.h>
```

```
using namespace
```

```
std; class Ass4
```

```
{
```

```
public:
```

```
    int N , i ,sum;
```

```
void get()
```

```
{
```

```
    cout<<"Input the Value for Nth Term --";
```

```

        cin>>N;
sum = 0;
    }
    void Series1()
    {
        cout<<"Series 1 IS --- \n\n";
for(i=1;i<=N;i++){          cout<<i<<" *
"<<i<<" = "<<(i*i)<<endl;
sum=sum+(i*i);
    }
        cout<<"\n\nSum of Above Series -- "<<sum<<"\n\n";
    }
    void Series2()
    {
        int s=0,j;
        cout<<"\nSeries 2 IS ---
\n\n";    for(i=1 ; i<=N ; i++){
for(j=1 , s=0 ; j<=i ; j++){
cout<<j<<" + ";
        s = s + j;
    }
        cout<<"\b\b= "<<s<<endl;
sum = sum + s;
    }
        cout<<"\n\nSum of Above Series -- "<<sum<<"\n\n";
    }
};

int main()
{

```



```

    system("cls");    Ass4 s1 ,s2;
cout<<"Program for Series Printing --
\n\n";    cout<<"1.Series 1 \t2.Series 2
\t3.Exit\n";    char c = getch();

    if(c == '1'){
s1.get();
s1.Series1();

    }

    else if (c == '2'){
s2.get();
s2.Series2();

    }

    else if (c == '3'){
exit(0);

    }

    cout<<"Do ypu want to Continue .. Press 1 Else any Key --
\n\n";

    c=getch();

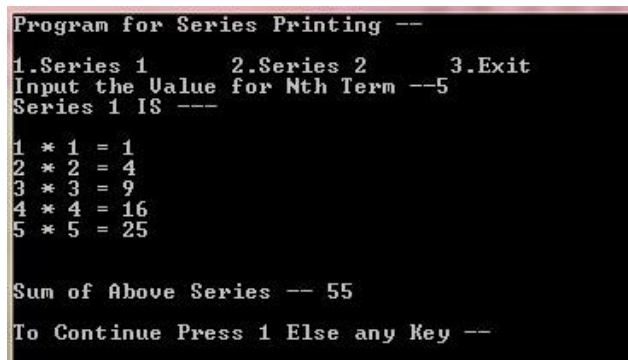
    if(c == '1'){
main();

    }

}

```

Pattern1



```

Program for Series Printing --
1.Series 1      2.Series 2      3.Exit
Input the Value for Nth Term --5
Series 1 IS ---
1 * 1 = 1
2 * 2 = 4
3 * 3 = 9
4 * 4 = 16
5 * 5 = 25

Sum of Above Series -- 55
To Continue Press 1 Else any Key --

```

Pattern2

```

Program for Series Printing --
1.Series 1      2.Series 2      3.Exit
Input the Value for Nth Term --5
Series 2 IS ---
1 = 1
1 + 2 = 3
1 + 2 + 3 = 6
1 + 2 + 3 + 4 = 10
1 + 2 + 3 + 4 + 5 = 15

Sum of Above Series -- 35
To Continue Press 1 Else any Key --

```

Q5.Write a code for input a string from user and print it in for following format. (For Eg: if you enter dog – Then output is d, do, dog, o , og, g). Solve this problem with String function or without function using class.

Code:

```

#include<iostream>

#include<string>

#include<algorithm>

using namespace std;

class Ass5
{
public:
    string s ;
    void get(string x){
        s = x ;
        cout<<"String is -- "<<s<<"\n\n";
    }
    void ShowWithfunction()
    {
        cout<<"\n\nShow With Function -- \n\n";
        for(int i = 0 ; i < s.length() ; i++){
            for(int j = 1 ; j <= s.length() - i ; j++){
                cout<<s.substr(i,j)<<" ";
            }
        }
    }
}

```

```

    }

    void ShowWithoutFunction()
    {
        cout<<"\n\nShow Without Function -- \n\n";

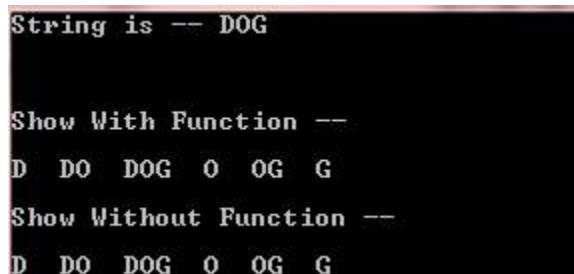
        string s1 = "";    for(int i =
0 ; i < s.length() ; i++){
            s1 = "";        for(int j = i ; j
< s.length() ; j++){            s1 = s1
+ s[j];            cout<<s1<<" ";
        }
    }
}

};

int main()
{
    Ass5 a ;
    a.get("DOG");
    a.ShowWithfunction();
    a.ShowWithoutFunction();
}

```

Output



```

String is -- DOG

Show With Function --
D DO DOG O OG G

Show Without Function --
D DO DOG O OG G

```

Q6. Write a C++ Code for input a multi word string from user and print into a particular format using function. Example: Hello World – Output – World Hello. Solve this problem with String function and without function using class.

Code:

```
#include<iostream>
```

```

#include<string>
#include<sstream>
using namespace std;
class Ass6
{
string s
;
public: void
get(string x){
    s = x ;
    cout<<"String is -- "<<s<<"\n\n";
}
void ShowReverseWordsWith()
{
    string s1 = "";
string out = "";
    cout<<"\n\nShow Words in Reverse Format With function -- \n\n";
    for(int i = 0 ; i < s.length() ; i++){
        if (s[i] == ' '){
s1 = s[i];          if (i ==
s.length()-1){
s1.append(" ");
out.insert(0,s1);
        }
        }else{(0,s1);
s1.clear();
        }
    }
}
}

```

```

        cout<<"After Reverse Operation String is -- \n\n";
cout<<out<<"\n\n";
    }

    void showReversewordsWithStringStream()
    {
        string s1 = "";
string out = "";

        cout<<"\n\nShow Words in Reverse Format With function -- \n\n";
        stringstream x(s);
while ( x >> s1 ){
out = s1 + " " + out ;
    }

        cout<<"After Reverse Operation String is -- \n\n";
cout<<out<<"\n\n";
    }

    void showReverseWordsWithout()
    {
        string s1="", out = "";
        cout<<"\n\nShow Words in Reverse Format Without Function -- \n\n";
        for(int i = 0 ; i < s.length() ;i++){
            if ( s[i]!=' '){
s1 = s1 + s[i];          if (i
== s.length()-1)
out = s1 + " " + out;
            }else{
                s1.append(" ");
out = s1 + out;
s1="";
            }

```

```
    }  
    cout<<"After Reverse Operation String is -- \n\n";  
cout<<out<<"\n\n";  
    }  
};
```

```
int main()  
{  
    Ass6 a;  
    a.get("Hello My Name is Jaishree");  
    a.ShowReverseWordsWith();  
    a.showReversewordsWithStringStream();  
    a.showReverseWordsWithout();  
}
```

Output

```
String is -- Hello My Name is Jaishree

Show Words in Reverse Format With function --
After Reverse Operation String is --
Jaishree is Name My Hello

Show Words in Reverse Format With function --
After Reverse Operation String is --
Jaishree is Name My Hello

Show Words in Reverse Format Without Function --
After Reverse Operation String is --
Jaishree is Name My Hello

Process returned 0 (0x0)    execution time : 0.454 s
Press any key to continue.
```

Q7. Write a C++ code for input a multi word string from user and print each word in reverse form and same position. For example: input – Hello World, Output – olleH dlroW.

Code:

```
#include<iostream>
#include<string>
#include<sstream>
#include<algorithm>
using namespace std;
class Ass7
{
public:
```

```

    string s ;
void get (string
x)
{
s = x;
    cout<<"String is -- "<<s;
}
void reverseStringOnPositionWithFun()
{
    string s1="",out="";
    cout<<"\n\nShow Data in Reverse on Position - \n\n";
    stringstream x(s);
while (x >> s1){
reverse(s1.begin(),s1.end());
out = out + s1 + " ";
    }
    cout<<"\n\nAfter Reverse Operation -- \n\n";
    cout<<out<<"\n\n";
}
void reverseStringOnPositionWithoutFun()
{
    string s1="",s2="",out="";
    cout<<"\n\nShow Data in Reverse on Position Without -
\n\n";
    stringstream x(s);
while (x >> s1){        for(int
i=0;i<s1.length();i++){
        s2 = s1[i] +s2 ;
    }
    out = out + s2 + " ";

```



```

        s2 = "";
    }

    cout<<"\n\nAfter Reverse Operation -- \n\n";
cout<<out<<"\n\n";

}

void reverseStringOnPositionWithoutFunPure()
{
    string s1="",s2="",out="";

    cout<<"\n\nShow Data in Reverse on Position Without Function- \n\n";

    for(int i=0;i<s.length();i++){

        if(s[i]!=' '){
s1 = s[i] +s1 ;           if(i
== s.length()-1){
out = out + s1;

        }

        }else {
out = out + s1 + " ";

        s1 = "";

        }

    }

    cout<<"\n\nAfter Reverse Operation -- \n\n";
cout<<out<<"\n\n";

}

};

int main()
{
    Ass7 a;

    a.get("Hello My Name is Jaishree");

    a.reverseStringOnPositionWithFun();

    a.reverseStringOnPositionWithoutFun();

```

```
a.reverseStringOnPositionWithoutFunPure();  
}
```

Output

```
After Reverse Operation --  
olleH yM emaN si eerhsiaJ  
  
Show Data in Reverse on Position Without -  
  
After Reverse Operation --  
olleH yM emaN si eerhsiaJ  
  
Show Data in Reverse on Position Without Function-  
  
After Reverse Operation --  
olleH yM emaN si eerhsiaJ  
  
Process returned 0 (0x0)    execution time : 0.394 s  
Press any key to continue.
```

Q8. Write a program to input 10 elements from user and find second maximum element?

Code:

```
#include<iostream>  
using namespace std;  
class ASS25  
{
```

```

public:
int
ARR[12];

int l , N;

ASS25(int x)
{
    N = x ;

    cout<<"\n\nPlease Enter Array Elements -- \n\n";

    for(l=0 ; l<N ; l++){
cout<<"Element ["<l+1<<" ] -- ";

        cin>>ARR[l];

    }

    show();

    SecondMax();
}

void show()
{
    cout<<"\nArray Element are -- \n\n";

    for(l=0 ; l<N ; l++){
cout<<ARR[l]<<" ";

    }

}

void SecondMax()
{
    int max1=0;

int max2=0;

for(l=0 ; l<N ; l++){

if(ARR[l] > max1){

```

```

max2 = max1;
max1 = ARR[I];
    }
    else if(ARR[I]<max1 && ARR[I]>max2){
        max2 = ARR[I];
    }
}
cout<<"\n\nSecond Maximum is --"<<max2;
}
};
int main()
{
int
N;
    cout<<"Enter Size of Array (1-10) -
- ";   cin>>N;   if(N>=1 && N<=10){
        ASS25 d(N);
    }
    else{
        cout<<"\n\nPlease Enter Valid Size...\n\n";
    }
}

```

Output

```

Enter Size of Array <1-10> -- 10

Please Enter Array Elements --

Element [1] -- 23
Element [2] -- 46
Element [3] -- 732
Element [4] -- 19
Element [5] -- 46
Element [6] -- 82
Element [7] -- 16
Element [8] -- 37
Element [9] -- 10
Element [10] -- 100

Array Element are --
23 46 732 19 46 82 16 37 10 100

Second Maximum is --100

```

Q9. Write a program to input 10 elements from user and insert in new array element of a given position by the user?

Code:

```

#include<iostream>

using namespace std;

class Ass26
{
public:
    int ARR[12];    int
    POS , DATA , I , N;

    Ass26(int x)
    {
        N = x ;

        cout<<"\n\nPlease Enter Array Elements -- \n\n";

        for(I=0 ; I<N ; I++){
            cout<<"Element ["<<I+1<<"] -- ";

            cin>>ARR[I];

        }

        show();

        cout<<"\nEnter New Element for Insert -- ";

        cin>>DATA;

        top:
    }

```

```

        cout<<"\nEnter Position between 0 to "<<N-1<<" -- ";
        cin>>POS;    if(!(POS>=0 &&
POS<=N-1)){        cout<<"\nEnter Valid
Position -- \n\n";

        goto top;
    }

    insertModule();
}

void show()
{
    cout<<"\nArray Element are -- \n\n";
    for(I=0 ; I<N ; I++){
cout<<ARR[I]<<" ";
    }
}

void insertModule()
{
    for(I=N ; I>POS ; I--){
        ARR[I]=ARR[I-1];
    }

    ARR[POS]=DATA;

N=N+1;

    cout<<"\n\nEnter Insertion of New Element -- ";
show();
}

};

int main()
{
    int
    N;

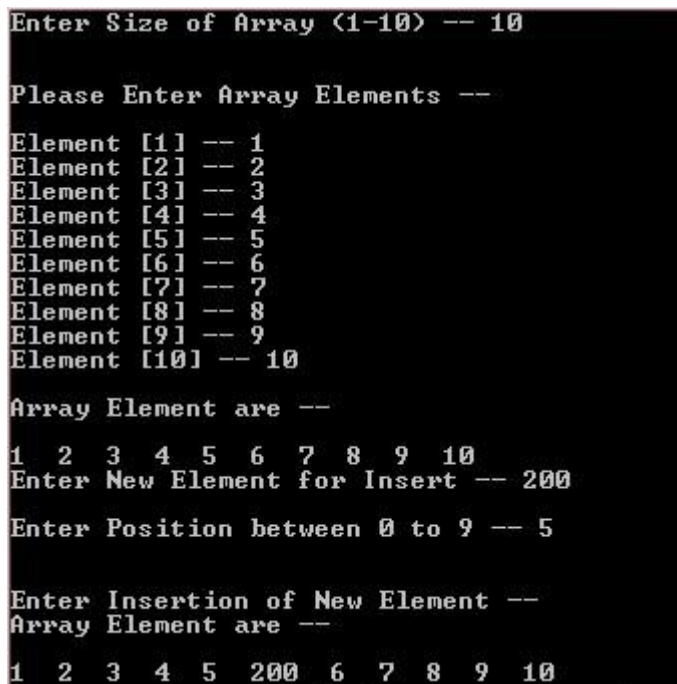
```

```

    cout<<"Enter Size of Array (1-10) -- ";
cin>>N;

    if(N>=1 && N<=10){
        Ass26 d(N);
    }
    else{
        cout<<"\n\nPlease Enter Valid Size...\n\n";
    }
}

```



The screenshot shows the output of a C++ program. It starts with a prompt 'Enter Size of Array (1-10) --' where the user enters '10'. Then it prompts 'Please Enter Array Elements --' and lists 10 elements from [1] to [10] with values 1 through 10. It then displays 'Array Element are --' followed by the sequence '1 2 3 4 5 6 7 8 9 10'. Next, it prompts 'Enter New Element for Insert --' where the user enters '200'. Then it prompts 'Enter Position between 0 to 9 --' where the user enters '5'. Finally, it prompts 'Enter Insertion of New Element --' and displays 'Array Element are --' followed by the sequence '1 2 3 4 5 200 6 7 8 9 10'.

```

Enter Size of Array (1-10) -- 10

Please Enter Array Elements --
Element [1] -- 1
Element [2] -- 2
Element [3] -- 3
Element [4] -- 4
Element [5] -- 5
Element [6] -- 6
Element [7] -- 7
Element [8] -- 8
Element [9] -- 9
Element [10] -- 10

Array Element are --
1 2 3 4 5 6 7 8 9 10
Enter New Element for Insert -- 200

Enter Position between 0 to 9 -- 5

Enter Insertion of New Element --
Array Element are --
1 2 3 4 5 200 6 7 8 9 10

```

Q10. WAP to input 10 elements from user and sort it in ascending order using bubble and selection sort.

```
#include<iostream> #include<conio.h> using namespace std;
```

```

class ass27
{
public:
    int ARR[12];

    int l , N ;

    ass27(int x)
    {
        N = x ;

        cout<<"\n\nPlease Enter Array Elements -- \n\n";

        for(l=0 ; l<N ; l++){
            cout<<"Element ["<<l+1<<"] -- ";

            cin>>ARR[l];

        }

        show();

        cout<<"\n\nChoose the Following Sort Types:-
\n";    cout<<"1.Selection Sort\n";

        cout<<"2.Bubble Sort\n";    cout<<"3.Exit\n";

        char c = getch();

        if (c == '1'){
            SelectionSort();
        }else if (c == '2'){
            BubbleSort();
        }else if (c == '3')
            exit(0);

        }

        void show()

        {

            cout<<"\nArray Element are -- \n\n";

            for(l=0 ; l<N ; l++){

                cout<<ARR[l]<<" ";

```



```

    }
}

void SelectionSort()
{
    int
i , j , t;

    for(i = 0 ; i < N ; i++){
for(j = i+1 ; j < N ; j++){
if(ARR[i] > ARR[j]){
t = ARR[i];

        ARR[i] = ARR[j];

        ARR[j] = t ;

    }

}

}

    cout<<"\nAfter Sorting Array Element are -- \n\n";

    for(l=0 ; l<N ; l++){
cout<<ARR[l]<<" ";

    }

}

void BubbleSort()
{
    int

i , j , t;

    for(i = 0 ; i < N-1 ; i++){
for(j = i+1 ; j < N-1 ; j++){
if(ARR[j] > ARR[j+1]){

        t = ARR[j];

        ARR[j] = ARR[j+1];

        ARR[j+1] = t ;

    }

}

}

}

```

```

        }
    }
    cout<<"\nAfter Sorting Array Element are -- \n\n";
    for(l=0 ; l<N ; l++){
cout<<ARR[l]<<" ";
        }
    }
};
int main()
{
    int
    N;
    cout<<"Enter Size of Array (1-10) -- ";
    cin>>N;
    if(N>=1 && N<=10){
        ass27 d(N);
    }
    else{
        cout<<"\n\nPlease Enter Valid Size...\n\n";
    }
}

```

```

Enter Size of Array <1-10> -- 10

Please Enter Array Elements --

Element [1] -- 21
Element [2] -- 31
Element [3] -- 61
Element [4] -- 54
Element [5] -- 71
Element [6] -- 81
Element [7] -- 19
Element [8] -- 43
Element [9] -- 69
Element [10] -- 100

Array Element are --

21 31 61 54 71 81 19 43 69 100

Choose the Following Sort Types:-
1.Selection Sort
2.Bubble Sort
3.Exit

After Sorting Array Element are --

19 21 31 43 54 61 69 71 81 100

```

Q11. Write a program to input two 3*3 matrix from user and multiply it? (Take all input from user, both arrays rows and columns and elements. Also check multiplication is possible or not.). using class.

```
#include<iostream>
```

```
using namespace std;
```

```
class Ass11 {    int A[3][3],
```

```
B[3][3], C[3][3];    int R1,
```

```
C1, R2, C2; public:
```

```
Ass11() {
```

```
    cout<<"----MATRIX MULTIPLICATION----"<<endl<<endl;
```

```
    cout<<"Enter rows of first matrix (0 - 3) - ";
```

```
    cin>>R1;    cout<<"Enter columns of first
```

```
matrix (0 - 3) - ";    cin>>C1;
```

```
    cout<<"Enter rows of second matrix (0 - 3) - ";
```

```
    cin>>R2;
```

```
    cout<<"Enter columns of second matrix (0 - 3) - ";
```

```
    cin>>C2;    if
```

```
(isMatrixCorrect()) {
```

```

cout<<endl;
getElements();
cout<<endl;
showArrays();
cout<<endl;
multiplyMatrix();
    }
}

bool isMatrixCorrect() {    return C1 == R2 && R1 <= 3 &&
R2 <= 3 && C1 <= 3 && C2 <= 3
    && R1 > 0 && R2 > 0 && C1 > 0 && C2 > 0;
}

void getElements() {
cout<<"First Array: ";
    cout<<"Enter element of a "<<R1<<"x"<<C1<<" array - \n";
    for (int i = 0; i < R1; i++){
for (int j = 0; j < C1; j++) {
        cout<<"["<<i+1<<"x"<<j+1<<"] - ";
        cin>>A[i][j];
    }
}

    cout<<"Second Array: ";
    cout<<"Enter element of a "<<R2<<"x"<<C2<<" array - \n";
    for (int i = 0; i < R2; i++){
for (int j = 0; j < C2; j++) {
        cout<<"["<<i+1<<"x"<<j+1<<"] - ";
        cin>>B[i][j];
    }
}
}

```

```

    }

    void showArrays() {
cout<<"First Array: "<<endl;

        for (int i = 0; i < R1; i++){
for (int j = 0; j < C1; j++) {
cout<<A[i][j]<<" ";

            }

            cout<<endl;

        }

        cout<<"Second Array: "<<endl;

        for (int i = 0; i < R2; i++){
for (int j = 0; j < C2; j++) {
cout<<B[i][j]<<" ";

            }

            cout<<endl;

        }

    }

    void multiplyMatrix() {
cout<<"Matrix Multiplication: "<<endl;

        for (int i = 0; i < R1; i++) {

            for (int j = 0; j < C2; j++) {
C[i][j] = 0;

                for (int k = 0; k < R2; k++){

                    C[i][j] += A[i][k] * B[k][j];

                }

                cout<<C[i][j]<<" ";

            }

            cout<<endl;

        }

    }

```

```

    }
};

int main() {

Ass11 a;

}

```

Output

```

----MATRIX MULTIPLICATION----
Enter rows of first matrix <0 - 3> - 3
Enter columns of first matrix <0 - 3> - 3
Enter rows of second matrix <0 - 3> - 3
Enter columns of second matrix <0 - 3> - 3

First Array: Enter element of a 3x3 array -
[1x1] - 1
[1x2] - 2
[1x3] - 3
[2x1] - 4
[2x2] - 5
[2x3] - 6
[3x1] - 7
[3x2] - 8
[3x3] - 9
Second Array: Enter element of a 3x3 array -
[1x1] - 1
[1x2] - 2
[1x3] - 3
[2x1] - 4
[2x2] - 5
[2x3] - 6
[3x1] - 7
[3x2] - 8
[3x3] - 9

First Array:
1 2 3
4 5 6
7 8 9
Second Array:
1 2 3
4 5 6
7 8 9

Matrix Multiplication:
30 36 42
66 81 96
102 126 150

```

Q12. Write a code for input a number and print it in for following format. (For Eg: if you enter 123 – Then output is 1, 12, 123, 2, 23,3).

Code:

```

#include<iostream
#include<string>

using namespace

std; class Ass33

{

public:

```

```

int n , i , j ;
string s;
void get()
{
    cout<<"Enter a Number -- ";
    cin>>n;
s=to_string(n);
}
void show()
{
    string t = "";
    cout<<"\nSeries is -- \n\n";
for(i=0;i<s.length();i++)
    {
        for(j=i;j<s.length();j++)
            {
t = t + s[j];
                cout<<t<<" ";
            }
        t = "";
    }
}
};
int main()
{
    Ass33 d;
    d.get();
    d.show();
}

```

Output

```
Enter a Number -- 1234
Series is --
1  12  123  1234  2  23  234  3  34  4
```

Q13. Write a program to enter a name from user and print star in the front of name alphabet on their alphabetic position vertically. Create code using class?

Code:

```
#include<iostream>
#include<string>
#include<iomanip>
using namespace std;
class Ass35
{
public:
    int n;
    string s;
    void get()
    {
        cout<<"Enter String -- ";
        getline(cin,s);
    }
    void show()
    {
        cout<<"Star in front alphabet on their position - \n\n";
        for(int i = 0 ; i<s.length();i++)
        {
            if(s[i]==' ')        continue;
            int p = toupper(s[i]-64);
            cout<<s[i]<<" - "<<setw(p)<<"*"<<endl;
        }
    }
}
```



```
};

int main()
{
    Ass35 d;

    d.get();

    d.show();
}
```

Output

```
Enter String -- jaishree
Star in front alphabet on their position -

j -          *
a -          *
i -          *
s -          *
h -          *
r -          *
e -          *
e -          *

Process returned 0 (0x0)   execution time : 6.533 s
Press any key to continue.
```

Q14. WAP to input a binary digit from user and print its decimal number using constructor.

Code:

```
#include<iostream>

#include<math.h>

using namespace
std; class Ass45
{
public:
    string s;
    int dec, p,
    c;
```

```

    Ass45(string x)
    {
s = x;

        dec = c = p = 0;

        for(int i = s.length()-1 ; i >= 0 ; i--
,c++){          p = pow(2,c);          dec =
dec + (((int)(s[i])-48) * p);

        }

        cout<<"Decimal of "<<s<<" is "<<dec<<endl;

    }
};

int main()
{
    string bin ;
    cout<<"Enter a binary Number -- ";
    cin>>bin;
    Ass45 d(bin);
}

```

Output

```

Enter a binary Number -- 10100
Decimal of 10100 is 20

```

Q15. WAP to input a four-digit number from user and multiply all prime digits available in number using single level inheritance with constructor.

Code:

```

#include<iostream
> #include<string>
using namespace
std; class Ass52
{
public:
    int n;

```

```

Ass52(int x)
{
    n = x;
}

};

class Derived : public Ass52
{
public:
    int mul , c , i ;
    Derived(int x1) : Ass52(x1)
    {
        if(to_string(n).length() != 4)
        {
            cout<<"Please Enter Valid Number..";
            return;
        }
        mul = 1;
        while (n > 0)
        {
            int x = n % 10 ;
            for(i = 2 ,c = 0 ;i <=x ; i++)
            {
                if (x % i == 0 ){
                    c++;
                }
            }
            if (c == 1)
            {
                mul =mul * x ;
            }
        }
    }
}

```

```

    }
    n = n / 10;
}
cout<<"Mul of Prime Numbers -- "<<mul<<"\n\n";
}
};
int main()
{
    cout<<"Enter a Number -- ";
    int n;
    cin>>n;
    Derived d(n);
}

```

Output

```

Enter a Number -- 1234
Mul of Prime Numbers -- 6

```

Q16. WAP for check password is perfect or not. (Password contain at least one Small & Capital alphabet, 1 symbol and 1 number, minimum 8 characters).

Code:

```

#include<iostream>
#include<string>
#include<conio.h>
using namespace
std;
int main()
{
    string pass = "Jaishree@1234";
    bool uc , lc , ss , pl , dig , sp;    uc
= lc = ss = pl = dig = sp = false;    pl =
(pass.length() >= 8) ? true : false;

```

```

for( int i = 0 ; i < pass.length() ; i++){
    if ( isupper(pass[i]))        uc =
true;    else if ( islower(pass[i]))
        lc = true;    else
    if ( isdigit(pass[i]))
        dig = true;    else if (
ispunct(pass[i]))
        ss = true;    else
    if ( isspace(pass[i]))
        sp = true;
    }
    if(uc && lc && ss && pl && dig && !sp){
cout<<"Password is Perfect";
    }
    else
        cout<<"Password is not Perfect";
}

```

output

Password is Perfect

Q17. Write a C++ program to sum of all positive integers in a sentence. Sample string: There are 12 chairs, 15 desks, 1 blackboard and 2 fans. Output: 30

Code:

```

#include<iostream>
#include<string>
#include<sstream>
using namespace std;
int main()
{

```

```

string s = "There are 12 chairs, 15 desks, 1 blackboard and 2
fans "; int n=0,sum =0; for(int i=0;i<s.length();i++){
    if(isdigit(s[i])){        n =
n * 10 + ((int)(s[i]))-48;
    }
    else {
        sum = sum + n;
        n=0;
    }
}
cout<<"Sum of all digit are -- "<<sum;
}

```

Output

```
Sum of all digit are -- 30
```

Q18. Write a C++ Program to input a number from user and perform swapping between even and odd position values. (Example: Input: 4159, Ouput: 1495, Input: 215, Ouput: 125)

Code:

```

#include<iostream
> #include<string>
using namespace
std; class Ass63
{
public:
    int n , t , l ,*arr;
void get()
{
    cout<<"Enter a Number - ";
    cin>>n;
}
}

```

```

        t = n ;

        l = to_string(n).length();
arr = new int[l];    for(int i = l-1 ;
i >= 0;i-- , t = t /10)    arr[i] =
t%10;

        t = 0 ;

    }

    void swapData()

    {

        for(int i = 0 ; i < l-1 ;
i+=2){            int x = arr[i];
arr[i] = arr[i+1];
arr[i+1] = x ;

        }

        for(int i = 0 ; i < l ; i++)

        {

            t = t * 10 + arr[i];

        }

    }

    void show()

    {

        cout<<"After Swapping of Index Elements Answer Will be --
\n\n";    cout<<"Before -- "<<n<<"\n\n";    cout<<"After --
"<<t<<"\n\n";

    }

};

int main()

{

    Ass63 d;

    d.get();

```

```

        d.swapData();
        d.show();
    }

```

Output

```

Enter a Number - 12345
After Swapping of Index Elements Answer Will be --
Before -- 12345
After -- 21435

```

Q19. WAP that can perform Factorial , Fibonacci and Tower of Hanoi operation using object oriented programming.

Code:

```

#include<iostream>

#include<conio.h>

using namespace
std; class Ass19
{
public:
    int Fact(int n)
    {
        if(n == 1)
            return 1;
        else
            return n * Fact(n-1);
    }

    int Fibo(int n)
    {
        if(n == 0)
            return 0;
        else if(n == 1){
            return 1;

```



```

    }
    else
        return Fibo(n-1)+Fibo(n-2);
}

void towerofhanoi(int n , char source , char auxiliary , char destination)
{
    if(n == 1){
        cout<<"Move disk 1 from "<<source<<" to "<<destination<<endl;
        return;
    }

    towerofhanoi(n-1,source , destination,auxiliary);    cout<<"Move
disk "<<n<<" from "<<source<<" to "<<destination<<endl;
    towerofhanoi(n-1,source,auxiliary,destination);

}

void MainMenu()
{
    system("cls");

    cout<<"Choose Any One Of The
Following:\n";    cout<<"1.Find Factorial\n";
    cout<<"2.Print Fibonacci\n";
    cout<<"3.Tower of Hanoi\n";
    cout<<"4.Exit\n\n";    char c = getch();

    if (c == '1'){
        cout<<"Find Factorial Module ";

        int n;

        cout<<"Enter a Number : ";

        cin>>n;

        cout<<"Factorial of "<<n<<" is ";

        cout<<Fact(n);

```

```

    }

    else if ( c == '2'){

        cout<<"Find Fibonacci Module ";

        int limit;

        cout<<"Enter a Limit : ";

cin>>limit;

        cout<<"Fibonacci Series -- \n";

        for(int i = 0 ; i < limit ; i++){

cout<<Fibo(i)<<" ";

        }

    }

    else if (c == '3'){

        cout<<"Tower of Hanoi Module..\n";

        cout<<"Enter limit(3-6) -- ";

        int n ;          cin>>n;

towerofhanoi(n,'A','B','C');

    }

    else if (c == '4'){

exit(0);

    }

    cout<<"\n\nDo you want to continue press 1 else any key ";

    c = getch();

if(c == '1')

    MainMenu();

    }

};

int main()

{

    Ass19 R;

```

```

    R.MainMenu();
}

```

Output

Menu

```

Choose Any One Of The Following:
1.Find Factorial
2.Print Fibonacci
3.Tower of Hanoi
4.Exit

```

Factorial

```

Choose Any One Of The Following:
1.Find Factorial
2.Print Fibonacci
3.Tower of Hanoi
4.Exit

Find Factorial Module Enter a Number : 5
Factorial of 5 is 120

Do you want to continue press 1 else any key

```

Fibonacci

```

Choose Any One Of The Following:
1.Find Factorial
2.Print Fibonacci
3.Tower of Hanoi
4.Exit

Find Fibonacci Module Enter a Limit : 10
Fibonacci Series --
0 1 1 2 3 5 8 13 21 34

Do you want to continue press 1 else any key

```

Tower of Hanoi

```

Choose Any One Of The Following:
1.Find Factorial
2.Print Fibonacci
3.Tower of Hanoi
4.Exit

Tower of Hanoi Module..
Enter limit(3-6) -- 4
Move disk 1 from A to B
Move disk 2 from A to C
Move disk 1 from A to C
Move disk 3 from A to B
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from A to B
Move disk 4 from A to C
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from A to B
Move disk 3 from A to C
Move disk 1 from A to B
Move disk 2 from A to C
Move disk 1 from A to C

Do you want to continue press 1 else any key

```

Q20. WAP to input size of Queue from user and create a queue using array. And perform push, pop and traverse operations using menu driven programming and functions.

Code:

```
#include<iostream>
#include<conio.h>
using namespace
std; class Queue
{
public:
    int QUEUE[10];
};
class QueueOperation : public Queue
{
public:
    int Front,Rear,MaxSize;
    QueueOperation()
    {
        Front = Rear = -1 ;
        cout<<"Enter Size of Queue -- ";
        cin>>MaxSize;
    }
    int Push()
    {
        system("cls");
        cout<<"Push Operation Module
\n\n";    int DATA , PushStatus=0;
        cout<<"Enter Data - ";
```

```

        cin>>DATA;    if(Front == 0 &&
Rear == MaxSize-1){
cout<<"Queue is Overflow...\n\n";

        return 0;

    }

    if(Front == -1 && Rear == -1){
Front = Rear = 0;

        QUEUE[Rear] = DATA;

        PushStatus = 1 ;

    }

    else if (Rear == MaxSize-1 && Front!= 0){

        for(int i = Front ; i < Rear ; i++){

            QUEUE[i-1] = QUEUE[i];

        }

        Front = Front -1 ;

        QUEUE[Rear] = DATA;

        PushStatus = 1;

    }

    else {

        Rear = Rear + 1;

        QUEUE[Rear] = DATA;

        PushStatus = 1;

    }

    if(PushStatus == 1)        cout<<"\n\nData
Successfully Inserted/Pushed..\n\n";

    }

    int Pop()

    {

        system("cls");

        cout<<"Pop From Queue Module \n\n";

```

```

        if(Front == -1 && Rear == -1){
cout<<"Queue is Underflow...\n\n";

        return 0;

    }

    if(Front == Rear){

        QUEUE[Front] = 0;

        Front = Rear = -1;

    }

    else {

        QUEUE[Front]= 0;

        Front = Front + 1;

    }

    cout<<"Data Deleted Successfully...\n\n";

}

int Peek()

{

    if(Front == -1){

        cout<<"Queue is Empty..\n\n";

        return 0;

    }

    else {

        for(int i = Front ; i
<= Rear ; i++){

cout<<QUEUE[i]<<" -> ";

        }

        cout<<"\n\n";

    }

}

void MainMenu()

{

```

```

        system("cls");
        cout<<"\tQueue Operation --
\n\n";    cout<<"1.Push ..\n";
cout<<"2.Pop ..\n";
cout<<"3.Peek ..\n";
cout<<"4.Exit .\n.";

        cout<<"Choose any one of these..\n\n";
        char c = getch();
        if(c == '1'){
            Push();
        }else if (c == '2'){
            Pop();
        }else if (c == '3'){
            Peek();
        } else if (c == '4'){
            exit(0);
        }

        cout<<"Do you want to continue press 1 else any key...\n";
        c = getch();
        if(c == '1'){
            MainMenu();
        }
    }
};

int main()
{
    QueueOperation Q;

    Q.MainMenu();
}

```

Output

Input Size

```
Enter Size of Queue -- 5
```

Menu

```
Queue Operation --  
1.Push..  
2.Pop..  
3.Peek..  
4.Exit..  
Choose any one of these..
```

Push

```
Push Operation Module  
Enter Data - 10  
  
Data Successfully Inserted/Pushed..  
Do you want to continue press 1 else any key...
```

Pop

```
Pop From Queue Module  
Data Deleted Successfully...  
Do you want to continue press 1 else any key...
```

Peek

```
Queue Operation --  
1.Push..  
2.Pop..  
3.Peek..  
4.Exit..  
Choose any one of these..  
  
Peek Module  
10 -> 20 ->  
Do you want to continue press 1 else any key...
```

Q21. WAP to input size of Circular Queue from user and create a circular queue using array. And perform push, pop and traverse operations using menu driven programming and functions.

Code:


```

#include<iostream>

#include<conio.h>

using namespace
std; class CQueue
{
public:
    int CQUEUE[10];

}

;

class CQueueOperation : public CQueue
{
public:
    int Front , Rear , MaxSize;

    CQueueOperation()
    {
        Front = Rear = -1;
    }

    cout<<"Enter Size of Queue -- ";
    cin>>MaxSize;

}

    int push()
    {
        system("cls");

        cout<<"Push in Queue Module..\n\n";

        int DATA , PS = 0;

        cout<<"Enter Element You Want to Insert -- ";

        cin>>DATA;        if ( Front == 0
&& Rear == MaxSize-1){
        cout<<"Queue is Overflow ..\n\n";

```

```

        return 0;
    }
    if ( Front == -1 && Rear == -1){
Front = Rear = 0;

        CQUEUE[Rear] = DATA;

        PS = 1;
    }
    else if (Rear == MaxSize-1 && Front != 0){
        Rear = 0;
        CQUEUE[Rear] = DATA;
        PS = 1;
    }
    else{
        Rear = Rear + 1;
        CQUEUE[Rear] = DATA;
        PS = 1;
    }
    if( PS == 1 )
        cout<<"\n\nData Successfully Inserted..\n\n";
}

int pop()
{
    system("cls");

    cout<<"POP From  Circular Queue Module..\n\n";

    if( Front == -1 && Rear == -1){
cout<<"Queue is Underflow ...\n\n";

        return 0;
    }

    if(Front == Rear){
        CQUEUE[Front] = 0;

```

```

        Front = Rear = -1;
    }
    else if( Front == MaxSize-1) {
        CQUEUE[Front] = 0;
        Front = 0;
    }
    else{
        CQUEUE[Front] = 0;
        Front = Front + 1;
    }
    cout<<"Pop Data Successfully\n\n";
}
int peek()
{
    if ( Front == -1){
        cout<<"\n\nQueue is Empty .. \n\n";
        return 0;
    }
    cout<<"Queue Element are -- \n\n";
    for(int POS = Front ; 1 ;){
        cout<<CQUEUE[POS]<<" -> ";
        if( POS == Rear){
            break;
        }
        POS = (POS == (MaxSize-1) && POS != Rear)?0:++POS;
    }
    cout<<"\n\n";
}
void MainModule()
{

```

```

        system("cls");

        cout<<"\t Circular Queue Operations --
\n\n";        cout<<"1. Push ..\n";
cout<<"2. POP ..\n";        cout<<"3. Peek ..\n";
cout<<"4. Exit ..\n";

        cout<<"Choose Any One of them ..\n";

        char c = getch();

        if(c == '1'){
push();
        }

        else if ( c == '2'){

            system("cls");

            pop();

        }

        else if( c == '3'){
system("cls");

            cout<<"Peek Module / Traversal..\n\n";

            peek();

        }

        else if(c == '4'){
exit(0);

        }

        cout<<"\n\nDo You Want to Continue .. Press 1 else any
key..";

        c =

getch();        if(
c == '1'){

            MainModule();

        }

    }
}

```

```
};

int main()
{
    CQueueOperation r;

    r.MainModule();    cout<<"\n\nThank
You for Using my APP..";
}Output
```

Input Size

```
Enter Size of Queue -- 5
```

Menu

```

Circular Queue Operation --
1.Push..
2.Pop..
3.Peek..
4.Exit..
Choose any one of these..
```

Push

```

Push Operation Module
Enter Data - 20

Data Successfully Inserted/Pushed..
Do you want to continue press 1 else any key...
```

Pop

```

POP From Circular Queue Module..
Pop Data Successfully

Do You Want to Continue .. Press 1 else any key..
```

Peek

```

Peek Module / Traversal..
Queue Element are --
10 -> 20 -> 30 ->

Do You Want to Continue .. Press 1 else any key..
```

Q22. WAP to input size of Stack from user and create a stack using array. And perform push, pop and peek operations using menu driven programming and functions.

CODE:

```
#include<iostream>

#include<conio.h>

using namespace
std;

class Stack
{
public:   int
STACK[10];
};

class StackOperation : public Stack
{
public:

    int TOP,MaxSize;

    StackOperation()
    {
        TOP = -1 ;
    }

    cout<<"Enter Size of Stack -- ";

    cin>>MaxSize;

    }

    int Push()
    {
        system("cls");
```

```

        cout<<"Push in Stack Module
\n\n";    int DATA , PushStatus=0;
cout<<"Enter Data - ";
cin>>DATA;    if(TOP == MaxSize-
1){        cout<<"Stack is
Overflow...\n\n";
        return 0;
    }
    if(TOP == -1){
        TOP = 0;
        STACK[TOP] = DATA;
        PushStatus = 1 ;
    }
    else {
        TOP = TOP + 1;
        STACK[TOP] = DATA;
        PushStatus = 1;
    }
    if(PushStatus == 1)        cout<<"\n\nData
Successfully Inserted/Pushed..\n\n";
}

int Pop()
{
    system("cls");
    cout<<"Pop From Stack Module \n\n";
    if(TOP== -1){        cout<<"Stack
is Underflow...\n\n";
        return 0;
    }
}

```

```

        STACK[TOP]=0;    TOP = TOP - 1 ;
cout<<"Data Deleted Successfully...\n\n";
    }
    int Peek()
    {
        cout<<"Peek Stack Module\n\n";
        if(TOP == -1){
            cout<<"Stack is Empty..\n\n";
            return 0;
        }
        else {
            for(int i =
TOP ; i >= 0 ; i--){
cout<<STACK[i]<<" -> ";
            }
            cout<<"\n\n";
        }
    }
    void MainMenu()
    {
        system("cls");    cout<<"\tStack
Operation -- \n\n";
        cout<<"1.Push..\n";
        cout<<"2.Pop..\n";
        cout<<"3.Peek..\n";
        cout<<"4.Exit..\n";    cout<<"Choose
any one of these..\n\n";
        char c = getch();
        if(c == '1'){
Push();

```



```

    }else if (c == '2'){
        Pop();
    }else if (c == '3'){
        Peek();
    } else if (c == '4'){
        exit(0);
    }

    cout<<"Do you want to continue press 1 else any key...\n";
    c = getch();
if(c == '1'){
    MainMenu();
}
}
};

int main()
{
    StackOperation S;
    S.MainMenu();
}

```

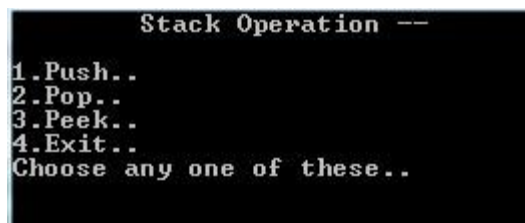
Output:

Input Size



```
Enter Size of Stack -- 5
```

Menu



```

Stack Operation --
1.Push..
2.Pop..
3.Peek..
4.Exit..
Choose any one of these..

```

Push

```
Push in Stack Module
Enter Data - 10

Data Successfully Inserted/Pushed..
Do you want to continue press 1 else any key...
```

Pop

```
Pop From Stack Module
Data Deleted Successfully...
Do you want to continue press 1 else any key...
```

Peek

```
Stack Operation --
1.Push..
2.Pop..
3.Peek..
4.Exit..
Choose any one of these..
Peek Stack Module
30 -> 20 -> 10 ->
Do you want to continue press 1 else any key...
```

Thank You

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